

Manipulating C-C coupling pathway in electrochemical CO₂ reduction for selective ethylene and ethanol production over single-atom alloy catalyst

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Shifu Wang^{1,2,9}, Fuhua Li^{3,4,9}, Jian Zhao², Yaqiong Zeng², Yifan Li⁵, Zih-Yi Lin⁶, Tsung-Ju Lee⁶, Shuhui Liu⁷, Xinyi Ren², Weijue Wang², Yusen Chen², Sung-Fu Hung⁶, Ying-Rui Lu⁸, Yi Cui⁵, Xiaofeng Yang^{6,2}, Xuning Li^{6,2}✉, Yanqiang Huang^{1,2}✉ & Bin Liu^{3,4}✉

Manipulation C-C coupling pathway is of great importance for selective CO₂ electroreduction but remain challenging. Herein, two model Cu-based catalysts, by modifying Cu nanowires with Ag nanoparticles (AgCu NW) and Ag single atoms (Ag₁Cu NW), respectively, are rationally designed for exploring the C-C coupling mechanisms in electrochemical CO₂ reduction reaction (CO₂RR). Compared to AgCu NW, the Ag₁Cu NW exhibits a more than 10-fold increase of C₂ selectivity in CO₂ reduction to ethanol, with ethanol-to-ethylene ratio increased from 0.41 over AgCu NW to 4.26 over Ag₁Cu NW. Via a variety of *operando*/in-situ techniques and theoretical calculation, the enhanced ethanol selectivity over Ag₁Cu NW is attributed to the promoted H₂O dissociation over the atomically dispersed Ag sites, which effectively accelerated *CO hydrogenation to form *CHO intermediate and facilitated asymmetric *CO-*CHO coupling over paired Cu atoms adjacent to single Ag atoms. Results of this work provide deep insight into the C-C coupling pathways towards target C₂₊ product and shed light on the rational design of efficient CO₂RR catalysts with paired active sites.

Electrochemical CO₂ reduction reaction (CO₂RR) offers a promising strategy to achieve carbon neutrality by converting CO₂ into value-added carbon-containing fuels or chemicals using renewable electricity^{1–5}. Compared to C₁ products, direct conversion of CO₂ to multi-carbon (C₂₊) products (e.g., ethylene, ethanol, etc.) is more

attractive^{6–10}. Unfortunately, the catalytic performance of electrochemical CO₂ reduction to C₂₊ products is still far from satisfactory with low selectivity and sluggish reaction kinetics^{11,12}.

The slow carbon-carbon (C-C) coupling reaction is the fundamental issue for poor conversion of CO₂ to C₂₊ products^{13,14}.

¹Department of Chemical Physics, University of Science and Technology of China, Hefei 230026, P. R. China. ²State Key Laboratory of Catalysis, Dalian Institute of Chemical Physics, Chinese Academy of Sciences, Dalian 116023, P. R. China. ³Department of Materials Science and Engineering, City University of Hong Kong, Kowloon, Hong Kong SAR 999077, China. ⁴Department of Chemistry, Hong Kong Institute of Clean Energy (HKICE) & Center of Super-Diamond and Advanced Films (COSDAF), City University of Hong Kong, Kowloon, Hong Kong SAR 999077, China. ⁵Vacuum Interconnected Nanotech Workstation, Suzhou Institute of NanoTech and NanoBionics, Chinese Academy of Sciences, Suzhou 215123, P. R. China. ⁶Department of Applied Chemistry and Center for Emergent Functional Matter Science, National Yang Ming Chiao Tung University, Hsinchu 300, Taiwan. ⁷School of Computer and Communication Engineering, Dalian Jiaotong University, Dalian 116028, P. R. China. ⁸National Synchrotron Radiation Research Center, Hsinchu 300, Taiwan. ⁹These authors contributed equally: Shifu Wang, Fuhua Li. ✉e-mail: lixn@dicp.ac.cn; yqhuang@dicp.ac.cn; blui48@cityu.edu.hk

Mechanistically, the CO₂-to-C₂₊ pathway (eg., ethylene, ethanol) is conventionally considered to proceed via a complex C-C coupling mechanism involving various intermediates such as *COCO, *COCHO, *COCO₂H, *CH₂CO, and *OCHCH₂, etc.^{1,15–25}. Therefore, regulating the key intermediates involved in the CO₂RR over catalytic sites and optimizing their binding energies become critical to adjust the C-C coupling pathways towards targeted C₂₊ products. Nevertheless, manipulation of C-C coupling pathways is still highly challenging due to the complexity of the catalytic system, the unclear reaction intermediates and the limited mechanistic understanding.

Cu-based materials have been well established as promising electrocatalysts capable to reduce CO₂ to multiple C₂₊ products^{11,26–28}. Till now, a large number of Cu-based tandem catalysts have been developed to realize high selectivity in CO₂RR to produce C₂₊ products, primarily resulting from the improved mass transport of *CO intermediate from CO-formation catalyst and the promoted C-C coupling over Cu surface^{29–32}. For instance, Cu-Ag catalyst often shows high activity for the formation of C₂₊ hydrocarbons and oxygenates^{33,34}. However, the tandem approach usually leads to uncontrolled C-C coupling, producing a variety of C₂₊ products such as ethylene, ethanol, and acetate through cascade catalysis. Grätzel et al.³⁵ and Yang et al.³⁴ suggested that the efficient *CO spillover from Ag to Cu could improve ethylene performance. Qiao et al.²³ claimed that significantly boosted ethanol production could be achieved by the tailored introduction of Ag to optimize the coordinated number and oxidation state of surface Cu sites. However, due to the complexity and diversity of the catalyst's structure, the underlying mechanism of CO₂RR to ethylene/ethanol in the Cu-Ag system remains unclear^{34,36,37}. Recently, the rapid development of single-atom catalysis has opened up new possibilities for constructing model catalysts with asymmetric catalytic sites^{38,39}. In particular, modification of copper-based catalysts with single metal atoms, and rational design of catalysts with dual- or multi- catalytic sites have been shown promising to induce asymmetric C-C coupling in CO₂RR^{40–42}. However, little understanding about the exact asymmetric C-C coupling mechanism is known.

In this work, via a modified galvanic replacement approach, two model Cu-based catalysts decorated with Ag nanoparticles (AgCu NW) and single Ag atoms (Ag₁Cu NW) were rationally designed and prepared for CO₂RR, which exhibited distinct selectivity in electrochemical CO₂ reduction to ethylene (Faradaic efficiency: ca. 54.9%, current density: ca. 156.0 mA cm⁻²) and ethanol (Faradaic efficiency: ca. 56.3%, current density: ca. 172.8 mA cm⁻²). Based on isotopic labelling experiments together with various *operando*/in-situ characterizations including *operando* X-ray absorption spectroscopy (XAS), *quasi* in-situ X-ray photoelectron spectroscopy (XPS), *operando* Raman spectroscopy and *operando* attenuated total reflectance surface enhanced infrared absorption spectroscopy (ATR-SEIRAS), the dynamic oxidation state evolution of Cu and Ag were discerned, and the key reaction intermediates involved in C-C coupling during CO₂RR, such as *COCO and *COCHO, were captured over AgCu NW and Ag₁Cu NW. Further coupled with density functional theory (DFT) calculations, the atomically dispersed Ag was identified to boost H₂O dissociation to release H⁺ and reduce the energy barrier for the formation of *CHO over its neighboring Cu site, leading to boosted asymmetric *CO-*CHO coupling over paired Cu atoms adjacent to the single Ag atom.

Results

Catalyst synthesis and characterization

The model Cu-based catalysts of Cu nanowires decorated with Ag nanoparticles and single Ag atoms were synthesized via a galvanic replacement strategy⁴³. In detail, the standard electrode potential difference drives the galvanic replacement reaction of 2Ag⁺(aq) + Cu(s) → 2Ag(s) + Cu²⁺(aq) (Supplementary Table 1). Figure 1a illustrates the synthesis of AgCu NW and Ag₁Cu NW and the experimental details are

summarized in the Supplementary Methods. One-dimensional Cu nanowires with an average diameter of 45 ± 3 nm were imaged by scanning electron microscopy (SEM) and transmission electron microscopy (TEM) (Fig. 1b and Supplementary Figs. 1, 2). As shown in Fig. 1c, the AgCu NW exhibits typical characteristics of supported catalyst, with Ag nanoparticles distributed as islands on the surface of the Cu NW (Supplementary Figs. 3, 4), and obvious grain boundaries between Ag nanoparticles and Cu NW can be clearly observed (Supplementary Fig. 4b). In contrast, Ag₁Cu NW displays similar microstructural features to Cu NW (Fig. 1d and Supplementary Figs. 5, 6). High-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM), energy-dispersive X-ray spectroscopy (EDX) elemental mapping and electron microscopy-based atom recognition statistics (EMARS) analysis on HAADF images of Ag₁Cu NW indicate that isolated Ag atoms are uniformly distributed on the surface of Cu NW matrix in Ag₁Cu NW (Fig. 1e and Supplementary Figs. 6c, d and 7)⁴⁴. The X-ray diffraction (XRD) patterns of Cu NW and Ag₁Cu NW exhibit identical diffraction peaks, which can be assigned to fcc Cu (Supplementary Fig. 8), indicating atomic dispersion of Ag atoms in Ag₁Cu NW⁴⁵. While for AgCu NW, diffraction peaks of both metallic Cu and Ag are clearly observable. The contents of Ag in Ag₁Cu NW and AgCu NW were measured by inductively coupled plasma emission spectrometer (ICP) and the results are summarized in Supplementary Table 2.

The surface valence states of the as-synthesized Cu NW, Ag₁Cu NW and AgCu NW were analyzed by X-ray photoelectron spectroscopy (XPS). The survey XPS spectra show a gradual increase of the Ag peak with increasing Ag content (Supplementary Fig. 9 and Supplementary Table 3). The high-resolution O 1s XPS spectrum of Cu NW, Ag₁Cu NW and AgCu NW exhibit similar characteristic peaks at 530.5 and 532.4 eV, which are related to adsorbed surface hydroxide (OH_{ads}) and adsorbed water (H₂O_{ads}), respectively (Supplementary Fig. 10)⁴⁶. As shown in Fig. 1f and Supplementary Fig. 11, the binding energy of Ag 3d displays a positive shift from 368.2 eV (Ag NW) to 368.4 (Ag₁Cu NW) and 368.3 eV (AgCu NW), implying a gradually decreased charge density on Ag due to electron transfer from Ag to Cu⁴⁷.

To further determine the chemical states and local coordination environments of Cu and Ag in Ag₁Cu NW and AgCu NW, X-ray absorption near-edge structure (XANES) and extended X-ray absorption fine structure (EXAFS) measurements were conducted. The Cu K-edge XANES spectra show that the edge features of Ag₁Cu NW and AgCu NW (designated as 1s → 4p transition) are close to those of the reference commercial Cu foil (Fig. 1g), indicating that Cu in Ag₁Cu NW and AgCu NW exist mainly in the form of Cu (0)⁴⁸. The corresponding EXAFS spectra of Ag₁Cu NW and AgCu NW also show the dominant Cu-Cu coordination at ca. 2.21 Å (Supplementary Figs. 12, 13 and Supplementary Table 4)⁴⁹. As displayed in Fig. 1h, the absorption edges of the Ag K-edge XANES for Ag₁Cu NW and AgCu NW are observed shifted towards higher energies compared to that for Ag foil, indicating a higher average valence state of Ag in Ag₁Cu NW and AgCu NW, matching well with the XPS results.

CO₂RR performance

The electrochemical CO₂RR performance of Cu NW, Ag₁Cu NW and AgCu NW were first evaluated in an H-type cell filled with CO₂-saturated 0.1 M KHCO₃ electrolyte (pH = 6.8), and the gas and liquid products were quantified by gas chromatography (GC) and nuclear magnetic resonance (NMR), respectively (Supplementary Figs. 14, 15). Linear sweep voltammetry (LSV) curves shown in Fig. 2a reveal that the current densities of both Ag₁Cu NW and AgCu NW outperform Cu NW in the potential range from -0.6 to -1.4 V vs. RHE. Meanwhile, the current densities of Cu NW, Ag₁Cu NW and AgCu NW in CO₂-saturated electrolyte are obviously higher than those in N₂-saturated electrolyte at the same applied potential, indicating that CO₂RR could occur readily compared to HER, consistent with the CV results (Supplementary Fig. 16). Figure 2b summarizes the potential-dependent

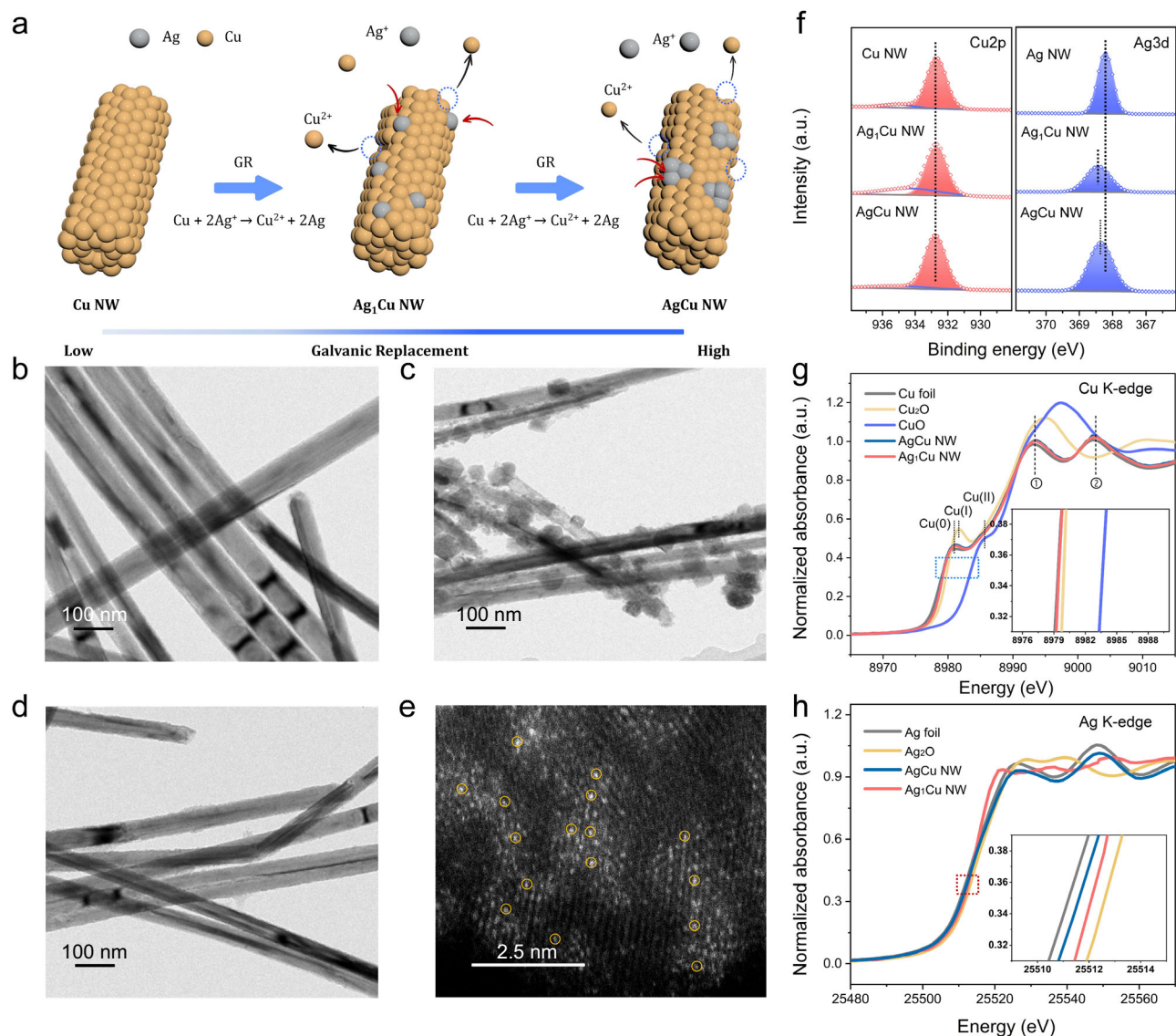


Fig. 1 | Synthesis and structural characterization. **a** Schematic illustration showing synthesis of Ag_1Cu NW and AgCu NW (Ag (gray) and Cu (yellow)). TEM images of **(b)** Cu NW, **(c)** AgCu NW and **(d)** Ag_1Cu NW. **e** HAADF-STEM image of Ag_1Cu NW. Notes: Ag atoms are not restricted to the positions indicated by the

yellow circles. **f** High-resolution Cu 2p and Ag 3d XPS spectra of Cu NW, Ag_1Cu NW, AgCu NW and AgCu NW. Normalized **(g)** Cu and **(h)** Ag K-edge XANES spectra of Ag_1Cu NW, AgCu NW and reference samples (inset shows the enlarged Cu and Ag K-edge XANES spectra).

Faradaic efficiencies (FEs) of different CO_2RR products over Ag_1Cu NW (I), AgCu NW (II) and Cu NW (III). Pure Cu NW electroreduced CO_2 to a wide variety of products including CO, HCOOH, C_2H_4 , and ethanol^{50,51}. The AgCu NW and Ag_1Cu NW exhibited greatly improved CO_2RR performance with distinctive selectivity towards ethylene and ethanol, with CO_2 -to-ethylene and CO_2 -to-ethanol Faradaic efficiency of ca. 42.4% and ca. 42.8%, respectively (Fig. 2b). The Faradaic efficiency of total C_2 products over Ag_1Cu NW could reach up to 65.1% at -1.22 V vs. RHE (Supplementary Figs. 17–19). Furthermore, the Ag_1Cu NW achieved the optimal ethanol Faradaic efficiency of 56.3% with a current density of 172.8 mA cm^{-2} at -1.0 V vs. RHE in a flow cell (Fig. 2c, Supplementary Figs. 20, 21a), while ethylene Faradaic efficiency reach a maximum value of 54.9% over the AgCu NW with a current density of 156.0 mA cm^{-2} at -1.1 V vs. RHE in a flow cell (Supplementary Figs. 21b, 22, 23 and Supplementary Table 5). Both Ag_1Cu NW and AgCu NW catalysts displayed outstanding catalytic stability, evidenced by the tiny decay in current density and ethanol/ethylene Faradaic efficiency during continuous electrolysis (Fig. 2d and Supplementary Fig. 24). The Ag_1Cu NW and AgCu NW exhibited similar surface structures,

valence states, and Ag species distributions after continuous electrolysis as observed in the fresh catalyst (Supplementary Figs. 25–27 and Supplementary Table 6).

Dynamic evolution of the catalytic sites under CO_2RR

To understand the vastly different CO_2 reduction to C_2 selectivity over AgCu NW and Ag_1Cu NW, *quasi* in-situ XPS was performed to investigate the surface properties of the Ag_1Cu NW and AgCu NW (Supplementary Fig. 28 and Supplementary Table 7). The *quasi* in-situ XPS spectra were collected at the open-circuit condition (OCV, immersed in KHCO_3 electrolyte), -0.7 and -1.2 V vs. RHE, respectively. As shown in Fig. 3a and Supplementary Fig. 29a, the high-resolution Cu 2p XPS spectrum of Ag_1Cu NW reveals the existence of partially oxidized Cu species under OCV condition. Only metallic Cu could be observed at -1.2 V vs. RHE, suggesting that the oxidized Cu species in Ag_1Cu NW could be completely reduced to metallic Cu during CO_2RR , which is also evidenced in the Cu Auger spectra collected before and after CO_2RR (Supplementary Fig. 30). Meanwhile, the oxidation state of Ag at -0.7 and -1.2 V vs. RHE was

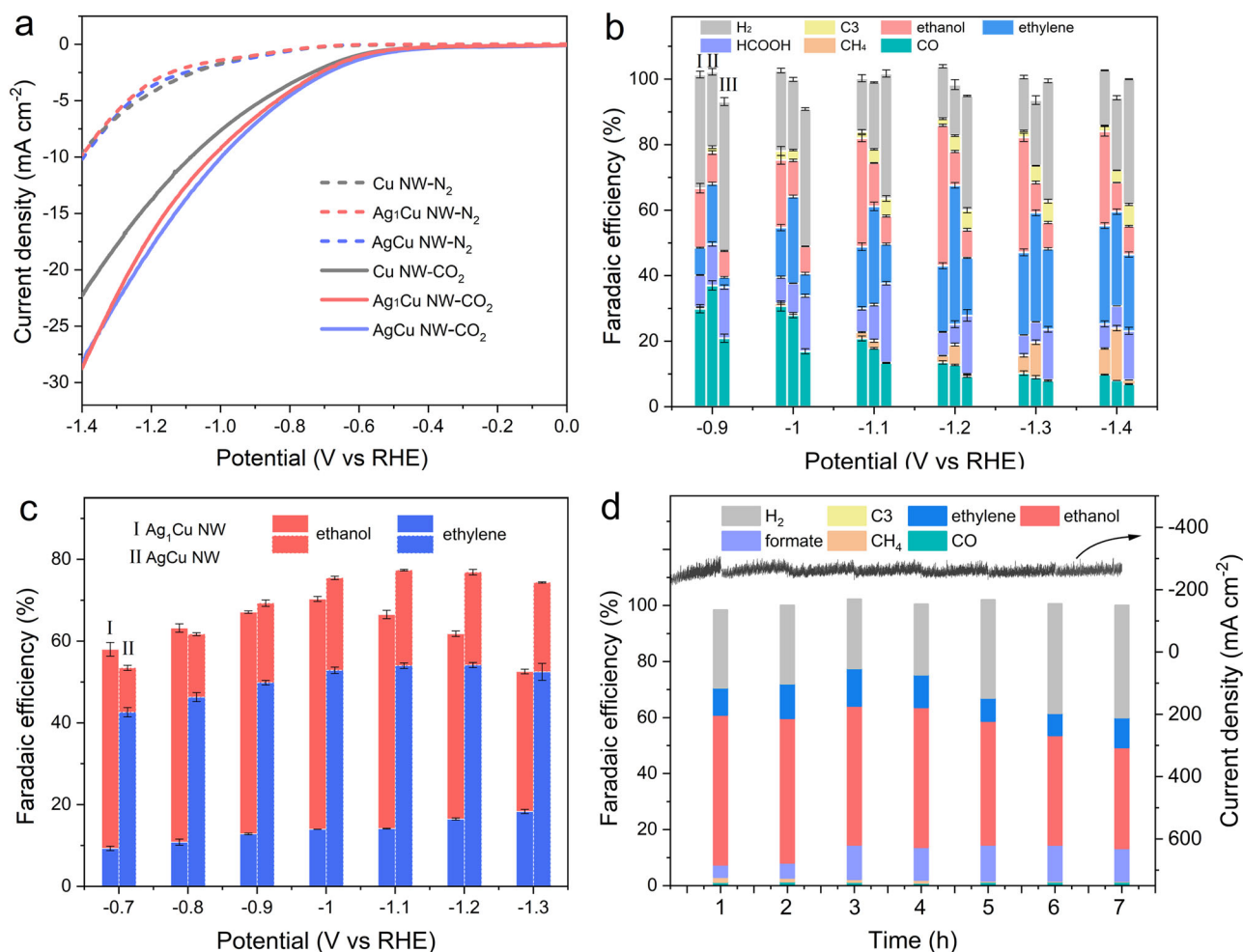


Fig. 2 | Electrochemical CO₂RR. a LSV curves acquired in CO₂- and N₂-saturated 0.1 M KHCO₃ solution on a rotating disc electrode at a rotation speed of 1600 rpm and a scan rate of 5 mV s⁻¹. **b** Potential dependent Faradaic efficiencies for CO₂RR on I: Ag₁Cu NW (left), II: AgCu NW (middle) and III: Cu NW (right) (H-type cell). **c** Faradaic efficiencies for CO₂-to-C₂ products on I: Ag₁Cu NW (left) and II: AgCu NW

(right) in 1 M KOH electrolyte (flow cell). Error bars represent the standard deviation of three independent measurements. All voltage was recorded without *iR* correction. **d** The stability test of Ag₁Cu NW acquired in a flow cell using 1 M KOH as the electrolyte at -1.0 V vs. RHE. The results of each potential in Fig. 2b, c were collected after one hour of continuous electrolysis.

observed to be higher than that under OCV condition (Fig. 3b and Supplementary Fig. 29b), which is possibly due to the adsorption of intermediate species on single Ag atoms during CO₂RR. As a reference, *quasi*-in-situ XPS spectra for Ag₁Cu NW were also collected in Ar atmosphere under identical conditions (Supplementary Fig. 31). The similar variation trend of Ag oxidation state as a function of applied potential suggests that the intermediate species bound to the single Ag atoms of Ag₁Cu NW may originate from water rather than CO₂. For AgCu NW, more negative shift in binding energies of both Cu and Ag were observed at the applied potential of -0.7 V and -1.2 V vs. RHE as compared to OCV condition, verifying that metallic Cu and Ag participated in the CO₂RR (Supplementary Figs. 32, 33).

To further investigate the dynamic evolution of catalytic sites over Ag₁Cu NW during CO₂RR, *operando* XAS measurements were conducted. As shown in Fig. 3c, *operando* XANES spectra of Cu K-edge were recorded at OCV, -0.7 V, -1.2 V vs. RHE, and AFT (after the CO₂RR), respectively. All spectra show characteristics of metallic Cu, displaying the similar pre-edge and white-line features (marked (1), (2), (3) in the XANES spectra of Fig. 3c) as those of Cu foil (Fig. 1g), revealing that the valence state of Cu in Ag₁Cu NW remained at Cu (0) throughout CO₂RR^{41,47}. At -0.7 V and -1.2 V vs. RHE, the Cu K-edge of Ag₁Cu NW moved towards lower energies as compared with that under OCV, which is due to the reduction of partially oxidized copper species.

The slight increase in Cu valence state of Ag₁Cu NW at AFT may result from desorption of CO₂RR intermediates from the Cu sites after removal of cathodic potential. The FT-EXAFS fitting profiles of the R space recorded at -0.7 V and -1.2 V vs. RHE indicated that the Cu-Cu coordination number of Ag₁Cu NW remained stable during the CO₂RR process, suggesting that Ag₁Cu NW maintained good stability throughout the reaction (Supplementary Figs. 34, 35 and Supplementary Table 8). On the contrary, the white-line energy of Ag in Ag₁Cu NW was observed gradually shifted to higher energies with increasing the applied cathodic potential from OCV to -1.2 V vs. RHE (Fig. 3d), further confirming the increase of valence state of Ag during CO₂RR, matching well with the *quasi*-in-situ XPS results. To extract the detailed structural information from EXAFS, we analyzed the Cu-Cu path at -0.7 and -1.2 V vs. RHE. The Cu-Cu coordination number (CN) of Ag₁Cu NW during the CO₂RR was quantified using the ARTEMIS program of IFEFFIT. The results indicated that the Cu-Cu coordination number of Ag₁Cu NW remained stable during the CO₂RR process, suggesting that the Ag₁Cu NW maintained good stability throughout the reaction.

In-situ capture of reaction intermediates

Operando Raman spectroscopy measurements were carried out to study the intermediate species over AgCu NW and Ag₁Cu NW involved in CO₂RR (Supplementary Fig. 36). Supplementary Table 9 summarizes

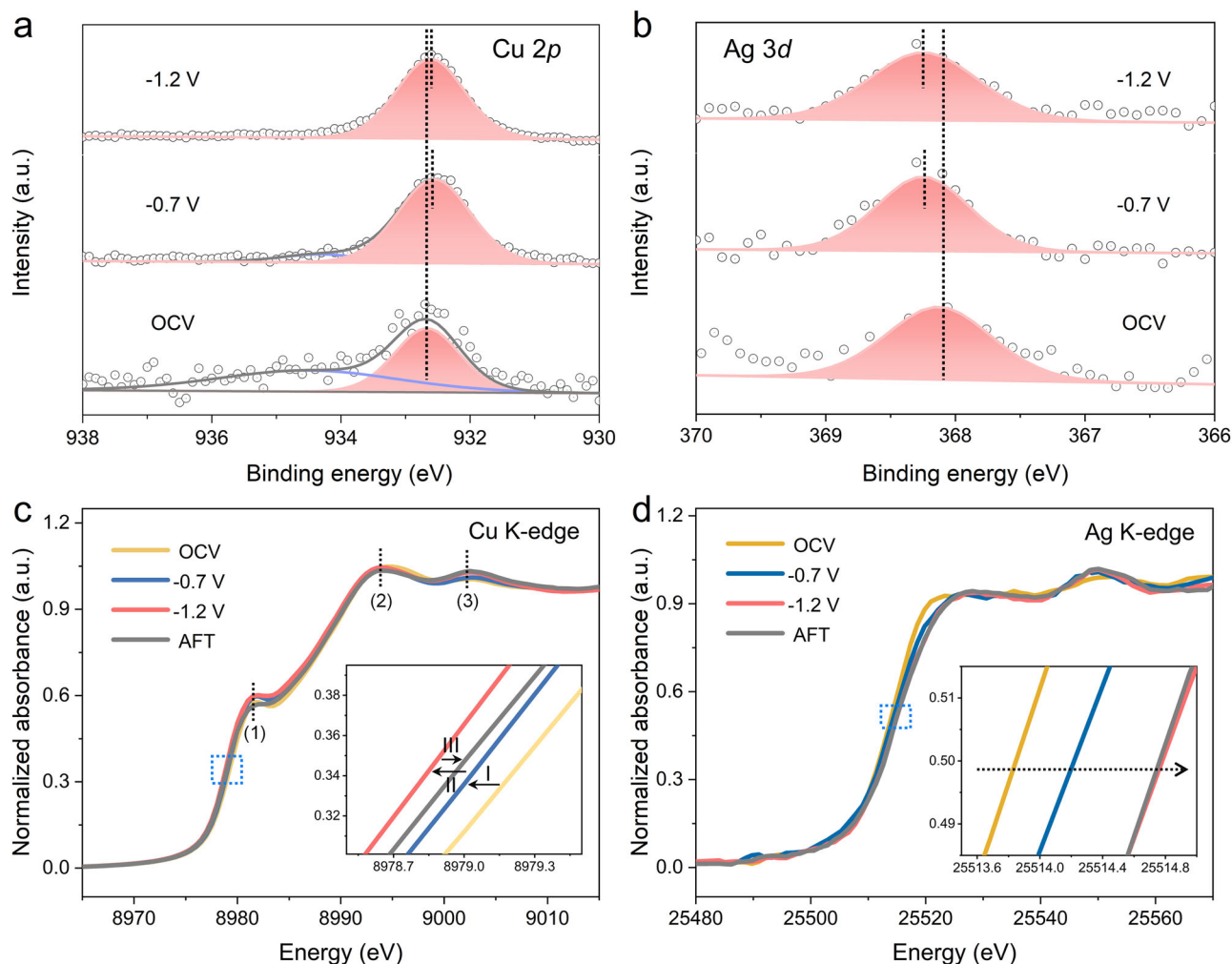


Fig. 3 | Dynamic evolution of Ag₁Cu NW during CO₂RR. a, b Quasi in-situ XPS spectra of high-resolution Cu 2p_{3/2} and Ag 3d_{5/2} for Ag₁Cu NW recorded at OCV, -0.7 V and -1.2 V vs. RHE in CO₂-saturated 0.1 M KHCO₃. **c, d** Normalized

operando Cu and Ag K-edge XANES spectra of Ag₁Cu NW recorded at OCV, -0.7 V, -1.2 V vs. RHE and AFT in CO₂-saturated 0.1 M KHCO₃ solution.

all Raman characteristic peaks observed over Cu NW, AgCu NW and Ag₁Cu NW. No signals corresponding to any intermediates could be captured in Ar-saturated 0.1 M KHCO₃ electrolyte (Supplementary Fig. 37). As shown in the *operando* Raman spectra recorded in CO₂-saturated electrolyte (Fig. 4a, b and Supplementary Figs. 38, 39), Raman peaks in the wavenumber range from 525 to 630 cm⁻¹ were observed over both Ag₁Cu NW and AgCu NW under OCV condition, which can be assigned to the Cu_xO species^{52–54}. With increase in applied cathodic potential, the Raman peaks of Cu_xO disappeared in all catalysts, attributed to reduction of Cu_xO to metallic Cu during CO₂RR. Over AgCu NW, three characteristic peaks associated with surface-adsorbed *CO were identified at 282, 362 and 1950–2150 cm⁻¹, corresponding to Cu-CO frustrated rotation (denoted as P1), Cu-CO stretching (denoted as P2) and C-O stretching, respectively⁵⁴. Figure 4c displays the potential-dependent intensity ratios between P2 and P1 over AgCu NW and Ag₁Cu NW. Compared to Ag₁Cu NW, the higher P2/P1 ratios observed over AgCu NW suggest higher *CO coverages on the AgCu NW surface, which shall benefit formation of *COCO intermediates for ethylene production^{35,55}. Besides, the Raman peaks at 704, 1072 and 1556 cm⁻¹ observed across the potential range from -0.6 to -1.4 V vs. RHE can be assigned to in-plane νCO₂, adsorbed carbonate species (νCO₃²⁻) and ν_{as}CO₂⁻, respectively⁵³. Over Ag₁Cu NW, a characteristic peak at 537 cm⁻¹ that can be assigned to *OH species, was observed at -0.6 V vs. RHE, which gradually grew with increasing

the applied cathodic potential (Fig. 4b and Supplementary Fig. 40)⁵⁶. The peak intensity of *OH species was found positively correlated with the Faradaic efficiency of ethanol, suggesting that the in-situ generated *OH species is crucial for the greatly promoted CO₂ reduction to ethanol over Ag₁Cu NW.

Operando ATR-SEIRAS measurements were further conducted to probe the reactive intermediates and provide mechanistic insights in CO₂ reduction to ethylene and ethanol over AgCu NW and Ag₁Cu NW (Supplementary Fig. 41). Figures 4d, e and Supplementary Figs. 42, 43 show the *operando* ATR-SEIRAS spectra recorded on AgCu NW, Ag₁Cu NW and Cu NW in CO₂-saturated 0.1 M KHCO₃. A distinct H₂O absorption peak at around 1620 cm⁻¹ was observed over all catalysts⁵⁷. As shown in Fig. 4d, two peaks appeared at around 2050 and 1865 cm⁻¹ over AgCu NW, which are associated with the atop-adsorbed *CO (*CO_{atop}) and bridge-adsorbed *CO (*CO_{bridge}) on Cu surface, respectively^{23,58,59}. The red-shift of the *CO peaks with increase in applied cathodic potential is due to the Stark effect arising from the decreased resonance frequencies in negative electric fields³⁸. In particular, the absorption peak of the *COCO intermediate at 1561 cm⁻¹ appeared at -0.4 V vs. RHE and reached the maximum intensity at -1.2 V vs. RHE, which is positively correlated with the ethylene Faradaic efficiency, suggesting that *COCO is crucial for ethylene generation over AgCu NW (Fig. 4d, f)¹². Additionally, a clear absorption peak could be identified at 1400 cm⁻¹ that can be assigned to the adsorbed *COOH

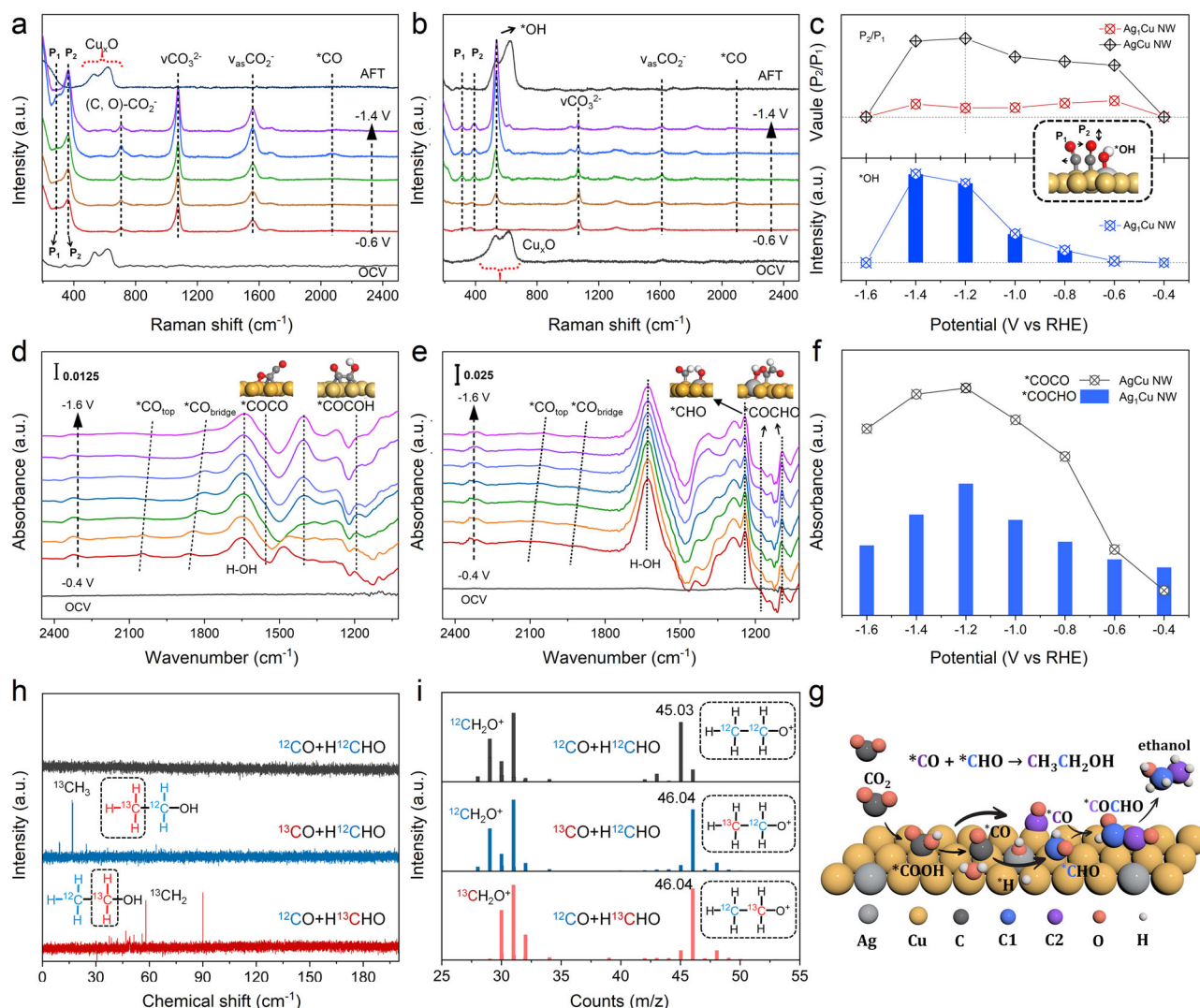


Fig. 4 | In-situ capture of reaction intermediates. Operando Raman spectra recorded in CO₂-saturated 0.1 M KHCO₃ solution at $E = -0.6$ V – -1.4 V vs. RHE over (a) AgCu NW and (b) Ag₁Cu NW, respectively. c Potential-dependent Raman peak intensity ratio of Cu–CO stretching (P₂) to Cu–CO frustrated rotation (P₁) over AgCu NW and Ag₁Cu NW, and peak intensity of *OH species versus potential over Ag₁Cu NW. d, e Operando ATR-SEIRAS spectra collected in CO₂-saturated 0.1 M KHCO₃ solution at $E = -0.4$ V – -1.6 V vs. RHE over AgCu NW and Ag₁Cu NW, respectively.

The inset in Fig. 4c, d and e shows the DFT-optimized structures. (Cu: yellow, Ag: gray, O: red, H: white, C: black gray) f Potential-dependent ATR-SEIRAS peak intensity for *COCHO intermediate over Ag₁Cu NW (blue bar) and *COCO intermediate over AgCu NW (black solid line). h ¹³C NMR and i MS spectra of CO₂RR products over Ag₁Cu NW, using ¹²CO + H¹²CHO (top), ¹³CO + H¹²CHO (middle), or ¹²CO + H¹³CHO (bottom) as the reactants, obtained at -1.2 V vs. RHE. g Schematic diagram showing the CO₂RR pathway over Ag₁Cu NW.

intermediate, whose intensity increased with cathodic potential rising from -0.4 to -1.6 V vs. RHE. The assignments for additional peaks observed in the ATR-SEIRAS spectra at -2332 , -1265 , -1183 and -1034 cm⁻¹ are summarized in Supplementary Table 10. Compared to AgCu NW, a blue-shift in *CO_{bridge} peak on Ag₁Cu NW was noticed at various applied potentials, suggesting enhanced binding strength of *CO on Ag₁Cu NW. Additional new peaks at around 1244 cm⁻¹ and 1083 cm⁻¹ were observed on Ag₁Cu NW, which can be assigned to adsorbed *CHO and *COCHO intermediates, respectively (Fig. 4e)⁶⁰. As shown in Fig. 4f, the peak intensity of *COCHO intermediate was found to gradually increase with increasing applied cathodic potential from -0.4 to -1.2 V vs. RHE, while the peak intensity of *CHO intermediate was correspondingly decreased, suggesting production of *COCHO intermediate from asymmetric *CO-*CHO coupling.

To probe the reaction pathway of CO₂ reduction to ethanol over Ag₁Cu NW, isotopic labelling experiments using ¹²CO + H¹²CHO, ¹²CO + H¹³CHO or ¹³CO + H¹²CHO as the reactants were carried out. As shown in Fig. 4h, ¹³C NMR signals were detected at chemical shifts of

17.47 and 58.05 , respectively, attributing to $-CH_3$ and $-CH_2OH$ from ethanol. Meanwhile, mass spectrometry (MS) spectra of liquid products from CO₂RR exhibit signals at $m/z = 45.03$ and $m/z = 46.03$, which can be attributed to ¹²CH₃¹²CH₂O⁺ and ¹³CH₃¹²CH₂O⁺ or ¹²CH₃¹³CH₂O⁺, respectively (Fig. 4i)⁶¹. Based on all these results, Fig. 4g schematically proposes the pathway of CO₂RR to ethanol over Ag₁Cu NW, where methyl C ($-CH_3$) and methylene C ($-CH_2OH$) in ethanol originate from *CO and *HCHO intermediates, respectively.

Theoretical insights into the catalytic mechanism

Density functional theory (DFT) calculations were performed to gain insights into the underlying reaction mechanisms for the distinctive selectivity of CO₂ electroreduction to ethylene and ethanol over AgCu NW and Ag₁Cu NW, respectively (Supplementary Figs 44, 45). Supplementary Figs. 46, 47 presents the optimized configurations of the reaction intermediates involved in CO₂RR over Ag₁Cu NW. As shown in Supplementary Figs. 48, 49, both the Bader- and Hirshfeld-based charge distribution results indicate that the atomically dispersed Ag in

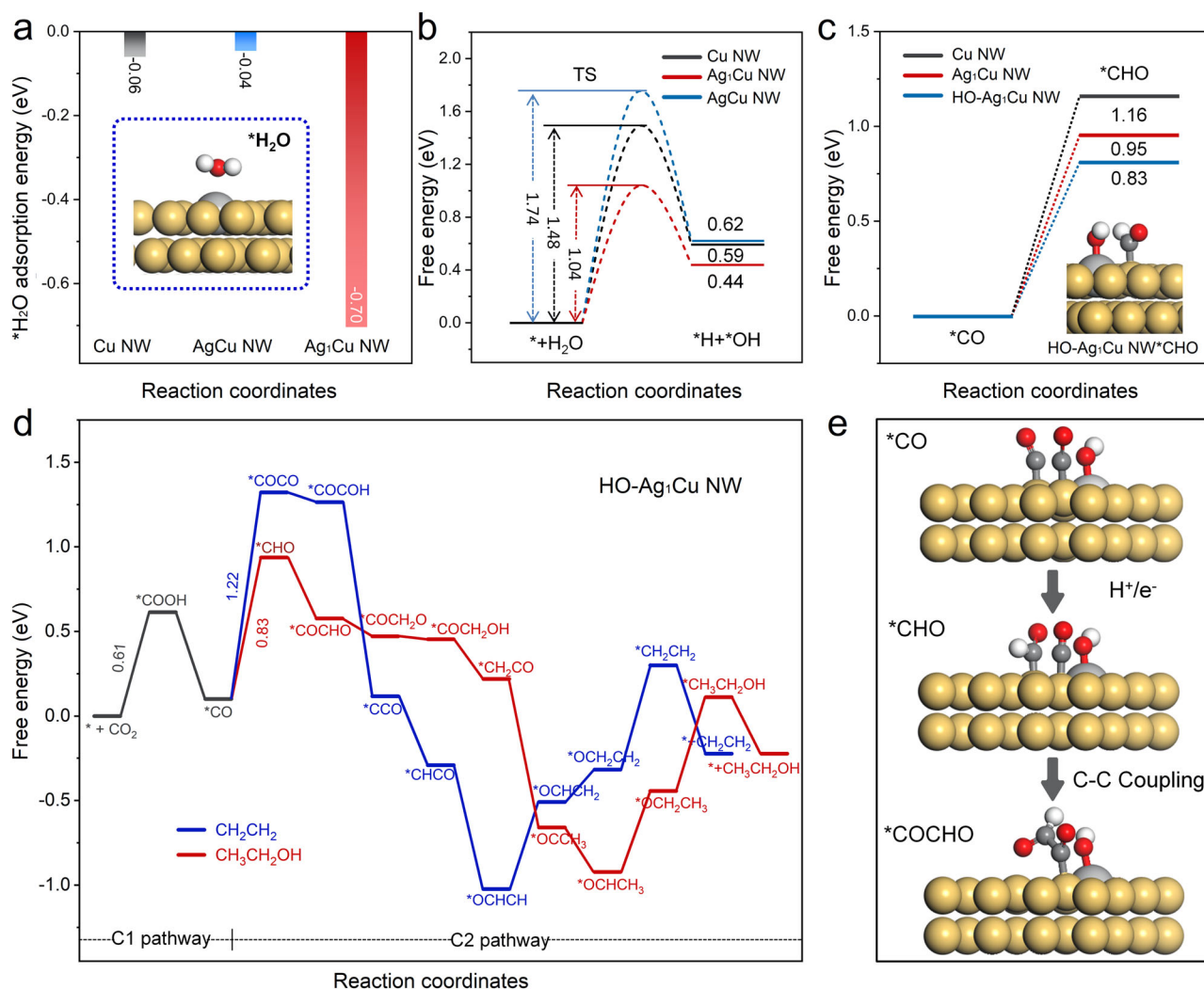


Fig. 5 | Theoretical insights into the catalytic mechanisms. a, b The adsorption and dissociation energies of H_2O on Cu NW, AgCu NW and Ag₁Cu NW. **c** Gibbs free energy diagrams for $^*\text{CO}$ hydrogenation to $^*\text{CHO}$ over Cu NW, Ag₁Cu NW and HO-Ag₁Cu NW. **d** Calculated Gibbs free energy diagrams for CO_2 RR to ethanol and

ethylene over Ag₁Cu NW. **e** Proposed reaction pathway for CO_2 reduction to ethanol over Ag₁Cu NW. The inset in Fig. 5a, c and e shows the DFT-optimized structures. (Cu: yellow, Ag: gray, O: red, H: white, C: black gray).

Ag₁Cu NW can donate electrons to its neighboring Cu atoms, moving the d-band center of Cu from -2.142 to -2.132 eV. Meanwhile, the atomically dispersed Ag atom becomes positively charged, matching well with the XPS and XAS results.

Operando Raman measurements observed in-situ generation of $^*\text{OH}$ over Ag₁Cu NW during CO_2 RR. To determine the $^*\text{OH}$ adsorption site, the adsorption and dissociation energies of H_2O on Cu NW, AgCu NW and Ag₁Cu NW were calculated¹⁷. As shown in Fig. 5a, compared to AgCu NW and Cu NW, the H_2O adsorption energy over Ag₁Cu NW is more negative with more electrons directly transferred from H_2O molecule to Ag₁Cu NW (Supplementary Fig. 50). Furthermore, H_2O tends to more preferably dissociate on the single Ag atomic site over Ag₁Cu NW with the lowest H_2O dissociation transition state (TS) energy barrier (Fig. 5b and Supplementary Figs. 51–53). The in-situ formation of HO-Ag₁Cu NW during CO_2 RR not only promoted the release of H^+ for the subsequent hydrogenation reaction, but also largely reduced the Gibbs free energy for $^*\text{CO}$ hydrogenation to $^*\text{CHO}$ (Fig. 5c), which facilitated the asymmetric $^*\text{CO}$ - $^*\text{CHO}$ coupling over the adjacent paired Cu atoms (HO-Ag₁-Cu-Cu) to preferably reduce CO_2 to ethanol (Fig. 5d, e, and Supplementary Figs. 54–57). On the other hand, for AgCu NW, as shown in Supplementary Figs. 58–60, the energy barrier for hydrogenation of $^*\text{CO}_2$ to $^*\text{COOH}$ can be greatly reduced from 0.83

to 0.63 eV, revealing preferred production of $^*\text{CO}$ via the $^*\text{COOH}$ route over AgCu NW. The increased $^*\text{CO}$ coverage shall increase the probability of $^*\text{CO}$ - $^*\text{CO}$ coupling to form $^*\text{COCO}$ intermediate, directing CO_2 reduction to produce ethylene over AgCu NW (Supplementary Figs. 61, 62).

Based on the above results, the reaction mechanisms of CO_2 RR for the distinctive selectivity towards ethylene and ethanol over AgCu NW and Ag₁Cu NW can be clearly evidenced. The ethylene pathway follows the symmetric C-C coupling, arising from the high $^*\text{CO}$ coverage generated over Ag nanoparticles and the reduced energy barrier for $^*\text{COCO}$ formation over AgCu NW. While the ethanol pathway follows the asymmetric C-C coupling, triggered by the in-situ generated HO-Ag₁-Cu-Cu sites (Fig. 5e), which reduces the energy barrier for the formation of $^*\text{CHO}$ species and thus promotes asymmetric $^*\text{CO}$ - $^*\text{CHO}$ coupling, boosting CO_2 reduction to ethanol over Ag₁Cu NW.

Discussion

In summary, two well-defined model Cu-based catalysts were developed by modifying Cu nanowires with Ag nanoparticles (AgCu NW) and single Ag atoms (Ag₁Cu NW), respectively, and their underlying C-C coupling mechanisms of CO_2 RR to ethylene and ethanol were successfully elucidated. *Operando* XAS and *quasi* in-situ XPS revealed

dynamic oxidation state evolution of Cu and Ag over AgCu NW and Ag₁Cu NW during CO₂RR. Following a combination of *operando* Raman spectroscopy and *operando* ATR-SEIRAS measurements, a higher *CO coverage was observed over AgCu NW surface, which would favor symmetric *CO-*CO coupling in CO₂RR to produce ethylene. On the contrary, the in-situ formed *OH species via H₂O dissociation was dominantly found over Ag₁Cu NW. Isotopic labelling experiments indicated that the methyl C (*CH₃) and methylene C (*CH₂OH) in ethanol product were originated from the *CO and *HCHO intermediates, respectively. Theoretical calculations further demonstrated that the atomically dispersed Ag could not only promote dissociation of H₂O to release H⁺, but also reduce the energy barrier for the formation of *CHO over its neighboring Cu site, resulting in the significantly promoted asymmetric *CO-*CHO coupling over paired Cu atoms adjacent to the single Ag atom, greatly boosting electrochemical CO₂ reduction to ethanol.

Methods

Chemicals and materials

Copper (II) chloride dihydrate (CuCl₂·2H₂O, 99.99%, CAS: 10125-13-0), glucose (C₆H₁₂O₆, 99.5%, CAS: 50-99-7), hexadecylamine (HAD, 98%, CAS: 143-27-1), hexane (98%, CAS: 110-94-3) and silver nitrate (AgNO₃, 99.99%, CAS: 7761-88-8) were purchased from Aladdin. Isopropyl alcohol (C₃H₈O, 99.9%, CAS: 67-63-0), potassium bicarbonate (KHCO₃, 99.95%, CAS: 298-14-6), potassium hydroxide (KOH, 99.99%, CAS: 1310-58-3), ethanol (CH₃CH₂OH, 99.9%, CAS: 64-17-5), and formaldehyde (HCHO, 20 wt. % in H₂O, 99 atom % ¹³C, CAS: 3228-27-1) were purchased from Sigma-Aldrich. Nafion solution (D521-1100ew, 5 wt.%, Dupont) was purchased from Alfa Aesar. Carbon paper (AVCarb P75, Product Code: 18070009; AVCarb P75T, 29.8–31.2% Wet Proofed, Product Code: 18070011) was purchased from FuelCell Store. All chemical reagents were used as received without any further purification. Deionized water (18.2 MΩ) was used in all experimental processes.

Preparation of Cu nanowires (Cu NW)

Uniform Cu NW were synthesized following a previously reported method with slight improvements⁶². In a typical synthesis, firstly, glucose (250 mg, 1.4 mmol), CuCl₂·2H₂O (117 mg, 0.7 mmol), and hexadecylamine (900 mg, 3.8 mmol) were dissolved in 50 mL of deionized water under vigorous stirring at 50 °C for 12 h. Subsequently, the homogeneous solution was kept in a water bath at 75 °C and stirred for 24 h until the solution turned brown. Afterwards, the above solution was transferred into a 150 mL pressure bottle and heated at 120 °C for 24 h. After cooling to room temperature, the products were collected by centrifugation at 16099.2 (×g) for 5 min. The obtained reddish-brown Cu NW was washed with hot deionized water (60 °C), ethanol, and hexane several times to remove excess hexadecylamine and glucose, and dried at 60 °C under vacuum for 12 h.

Synthesis of AgCu NW and Ag₁Cu NW

AgCu NW and Ag₁Cu NW were synthesized via a modified galvanic replacement approach⁶³. Firstly, Cu NW (32 mg, 0.5 mmol) was dispersed in 50 mL of isopropyl alcohol in a three-necked flask under sonication for 30 min to form a reddish-brown suspension. Immediately afterwards, a certain amount of AgNO₃ isopropyl alcohol solution was slowly injected into the above Cu NW suspension via a peristaltic pump at a rate of 0.5 mL min⁻¹ to trigger the galvanic replacement reaction. The added AgNO₃ are 0.1 and 0.016 mmol to prepare AgCu NW and Ag₁Cu NW. Then, the mixed suspension was kept at room temperature for 30 min with continuous magnetic stirring. The resulting red-grey product was collected by centrifugation at 8000 rpm for 5 min and washed three times with deionized water and ethanol, and dried at 60 °C under vacuum for 12 h. It should be noted that the entire reaction process required Ar as the protective gas to avoid oxidation.

Characterization

The crystal structure of Cu NW, AgCu NW and Ag₁Cu NW were characterized by X-ray diffraction (XRD, Bruker AXS D8 Advance) with Cu K_α radiation (λ = 1.5406 Å). The morphology of the catalysts was examined by transmission electron microscopy (HRTEM, JEOL JEM-2100F operated at 200 kV) and field-emission scanning electron microscopy (FESEM, JSM-6700F) equipped with an energy dispersive X-ray spectroscopy (EDX, Aztec X-Max 80 T). High-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) characterization was conducted on a JEOL JEMARM200F STEM/TEM with a guaranteed resolution of 0.08 nm. Detailed chemical compositions were analyzed by X-ray photoelectron spectroscopy (XPS) on an ESCALAB 250 photoelectron spectrometer (Thermo Fisher Scientific) using a monochromatic Al K_α X-ray beam (1486.6 eV). All binding energies were referenced to the C1s peak (284.8 eV). The concentration of Ag in AgCu NW and Ag₁Cu NW were quantified by inductively coupled plasma optical emission spectroscopy (ICP-OES). For ex-situ XAS measurements, a certain amount of powder samples were placed into a hollow circular mold (made of polytetrafluoroethylene) with a diameter of 1 cm. The filled sample was then fixed and vacuumed to prevent oxidation during the transfer and measurement processes. And the sample was secured on an XAS sample stage, which was positioned at a 45° angle to the X-ray during the measurement, allowing for collection of fluorescence signals.

Quasi in-situ XPS measurements

Quasi in-situ X-ray photoelectron spectroscopy characterizations (XPS) were conducted at the Vacuum Interconnected Nanotech Workstation (Nano-X), Suzhou Institute of Nano-Tech, and Nano-Bionics, Chinese Academy of Sciences (CAS). Quasi in-situ XPS measurements were carried out in a SPECS PHOIBOS 100 analyzer using monochromatic X-ray source of Al K_α radiation (1486.6 eV), and the overall spectrum resolution is Ag 3d_{5/2}, <0.5 eV FWHM at 20 kcps@UHV. The working electrode (1 × 1 cm²), consisting of carbon paper with drop-casted catalyst, was immersed in CO₂-saturated 0.1 M KHCO₃ solution. To avoid surface oxidation of the catalyst, the H-type cell was placed in a glove box for the CO₂RR. After CO₂RR, the sample was rapidly transferred through a chamber with ultrahigh vacuum for XPS testing. Quasi in-situ XPS spectra were collected at open-circuit condition (OCV, immersed in KHCO₃ electrolyte), -0.7 and -1.2 V vs. RHE, respectively.

Operando XAS measurements

X-ray Absorption Spectroscopy (XAS) and operando XAS measurements were performed in the total-fluorescence yield (TFY) mode at room temperature. These measurements utilized the TPS 32 A and TLS01C1 beamlines at the National Synchrotron Radiation Research Center (NSRRC) in Taiwan. Energy calibration was performed with a Cu and Ag foil standard by shifting all spectra to a glitch in the incident intensity. Fluorescence spectra were recorded using a silicon drift detector. The measurement in a typical three-electrode setup, the same condition as that in electrochemical characterization was performed in a specially designed Teflon container with a window sealed by Kapton tape^{64–66}. CO₂RR was conducted on a Bio-logic VSP potentiostat in a standard three-electrode setup. The spectral time recording was 40 min for a single spectrum (2 s for the pre-edge region, 2 s for the rising-edge region, and 8 s for the EXAFS region) during CO₂RR. Carbon paper (1 × 1 cm²) supported catalyst as the working electrode was used for operando XAS measurements. To prepare the catalyst ink, catalyst (5 mg) and 5 wt.% Nafion solution (40 μL) were introduced into water and isopropyl alcohol solution (960 μL, 1:1 v/v) and ultrasonicated for 3 h. Next, the catalyst ink (400 μL) was applied onto a carbon paper and allowed to dry in air, affording a catalyst loading of 2 mg cm⁻². The pre-edge baseline was subtracted, and the spectrum was normalized to the post-edge. EXAFS analysis was conducted using

a Fourier transform on k^2 -weighted EXAFS oscillations to evaluate the contribution of each bond pair to the Fourier transform peak.

Operando Raman measurements

Operando Raman spectra were recorded at controlled electrochemical potentials in CO_2 -saturated 0.1 M KHCO_3 electrolyte in a three-electrode polytetrafluoroethylene cell with Pt wire as the counter electrode and Ag/AgCl as the reference electrode. Operando Raman spectra were collected on a Raman spectrometer (Bruker NanoWizard Ultra Speed & inVia Raman) with a 532 nm/785 nm excitation laser (5%), having a power of 2.1 mW measured at the objective. A 50 $\times$ long working distance objective (Olympus, 0.5 NA) was used, focusing on the sample surface and avoiding contact to the electrolyte. Acquisition time was set as 120 s for the spectral Raman shift ranging from 200 to 3000 cm^{-1} . Operando Raman experiments were performed on a carbon paper as the working electrode with a catalyst loading of 0.5 mg cm^{-2} . CO_2 -saturated electrolyte was prepared by purging CO_2 (99.99%) into 0.1 M KHCO_3 aqueous solution for 30 min, and a flow of CO_2 was maintained over the electrolyte throughout electrochemical measurements.

Operando ATR-SEIRAS measurements

The operando attenuated total reflectance surface enhanced infrared absorption spectroscopy (ATR-SEIRAS) measurements were performed on a Nicolet iS20 FTIR spectrometer equipped with a MCT detector cooled using liquid nitrogen and PIKE VeeMAX III variable angle ATR sampling accessory. The catalyst was added dropwisely to the surface of Au-plated silicon as the working electrode. A graphite rod and an Ag/AgCl electrode were used as the counter and reference electrode in all tests, respectively. The background was taken at OCV in CO_2 -saturated 0.1 M KHCO_3 electrolyte. Afterwards, potential-dependent spectra were collected during stepping the working electrode potential from -0.4 V to -1.6 V vs. RHE. The step width and duration are 0.1 V and 120 s, respectively.

Electrochemical measurements

H-type cell. Electrochemical measurements were performed at room temperature and ambient pressure on a glassy carbon electrode in a two-compartment H-cell filled with CO_2 saturated 0.1 M potassium bicarbonate (KHCO_3) electrolyte using a CHI660e electrochemical workstation. A three-electrode cell configuration was employed, with a glassy carbon rotating disc electrode (RDE) (EE300a1 Rotating Disk Electrode, diameter: 5 mm) as the working electrode, a platinum foil ($1.5 \times 1.5 \text{ cm}^2$) as the counter electrode, and a saturated calomel electrode as the reference electrode (SCE, $\text{Hg/Hg}_2\text{Cl}_2$). To ensure the accuracy of the RHE, we periodically calibrated the reference electrode in H_2 -saturated 0.1 M HClO_4 and 0.1 M KOH electrolytes before conducting experiments. A reversible HOR/HER reaction was performed using two Pt plates as both the working and counter electrodes, with H_2 bubbling over the working electrode during the process. For calibration, the clean Pt electrode was cycled at a slow sweep rate of 10 mV s^{-1} around the expected RHE value in each electrolyte until the CV curve stabilized. The RHE potentials in the two electrolytes were then determined from the corresponding open-circuit potentials. H-cell separated by a proton exchange membrane (Nafion 117, 183 μm , Dupont). And the Nafion membrane needs to be pretreated. The steps are as follows: (1) Place the cut Nafion 117 membrane in a beaker, add an excess of 5 wt% H_2O_2 solution, and boil it in a water bath at 80°C for 0.5 h to remove organic and inorganic impurities from the membrane and rinse it 2–3 times with deionized water. (2) Add 0.5 mol L^{-1} HNO_3 solution and boil it in a water bath at 80°C for 0.5 h and rinse it 2–3 times with deionized water. The same three-electrode system, but with carbon paper ($1 \times 1 \text{ cm}^2$) as the working electrode, was used for stability test. For all the measurements, CO_2 (with 1% Ar) was continuously purged

into the solution at a flow rate of 20 SCCM by a mass flow controller (CS200A, Beijing Sevenstar Flow Co., LTD). Cyclic voltammetry (CV) and linear sweep voltammetry (LSV) acquired in CO_2 -saturated 0.1 M KHCO_3 solution on a RDE at a rotating speed of 1600 rpm with a scan rate of 5 mV s^{-1} . All potentials were calculated with respect to the reversible hydrogen electrode (RHE) scale according to the Nernst equation: $E(\text{vs. RHE}) = E(\text{vs. SCE}) + 0.2415 + 0.0591 \times \text{pH}$. The voltage was recorded without iR correction. The pH was determined by a pH meter (S220 SevenCompactTM pH/Ion). Supplementary Fig. 63 shows the ohmic drop during CO_2 RR. To prepare the working electrode, catalyst (5 mg) and 5 wt% Nafion solution (40 μL) were introduced into water and isopropyl alcohol solution (960 μL , volume ratio of 1:1) and ultrasonicated for 3 h. Next, the catalyst ink (20 μL) was applied onto a glassy carbon RDE (diameter: 5 mm) and allowed to dry in air, affording a catalyst loading of 0.5 mg cm^{-2} . For stability test, catalyst ink (100 μL) was brushed onto carbon paper (AVCarb P75T, 29.8–31.2% Wet Proofed, Product Code: 18070011, Purchased from FuelCellStore, $1 \times 1 \text{ cm}^2$) at 60°C , affording a catalyst loading of 0.5 mg cm^{-2} .

Flow cell. The electrochemical measurements were performed in a flow cell (manufactured by Dioxide Materials Company) separated by an anion exchange membrane (FAA-3-PK-130, Fumasep, Soak the membrane in 1 M KOH for 24 h before use.) using a CHI660E electrochemical workstation at room temperature. The flow cell was equipped with three electrodes, in which the Ag/AgCl electrode as the reference electrode, Ni-foam (200–250 μm , Youveim) as the counter electrode and the obtained GDE as the working electrode. CV and LSV curves were recorded on a CHI660E electrochemical workstation at a scan rate of 5 mV s^{-1} from -1.0 V to -2.5 V vs. SCE. 1 M KOH electrolyte was used as the anolyte and was circulated using a peristaltic pump (F01A-STP, Kameor) at 2.5 mL min^{-1} . The flow rate of the CO_2 gas flowing into the cathode flow field was kept at 20 SCCM by a mass flow controller (CS200A, Beijing Sevenstar Flow Co., LTD). All potentials were converted to the reversible hydrogen electrode (RHE) scale by the following equation: $E(\text{vs. RHE}) = E(\text{vs. Ag/AgCl}) + 0.197 \text{ V} + 0.0591 \text{ pH}$. The voltage was recorded without iR correction.

Preparation of gas diffusion layer (GDL). Carbon paper (AVCarb P75, Product Code: 18070009, Purchased from FuelCellStore) was repeatedly soaked in a polytetrafluoroethylene (PTFE, DISP 40LX, CAS: 25667-42-9) solution until the weight of PTFE reached 25 wt%. The carbon paper was heated in air at 350°C for 2 h. A carbon black (Vulcan XC-72R, Sigma) ink containing 40 wt% PTFE was drop-casted on one side of the above carbon paper. After further heating in air at 350°C for 2 h, the GDL was prepared.

Preparation of gas diffusion electrode (GDE). The catalyst (4 mg) and Nafion solution (DuPont, ~5 wt% in a mixture of isopropyl alcohol and deionized water) were firstly mixed and dispersed into isopropyl alcohol (1 mL) under ultrasonication for 30 min to form a uniform catalyst ink. Then, the catalyst ink (1 mL) was spray-casted onto a GDL ($2 \times 2 \text{ cm}^2$), affording a catalyst loading of 1 mg cm^{-2} , and dried at 60°C to form the GDE.

Faradaic efficiency evaluation

The products of electrochemical CO_2 reduction reaction (CO_2 RR) were measured using chronoamperometry at each fixed potential in a CO_2 RR electrolysis cell. The electrolyte in the cathodic compartment was stirred at a rate of 500 rpm during electrolysis. CO_2 gas was introduced into the cathodic compartment at a rate of 20 SCCM and was routed into a gas chromatograph (GC, Agilent 7890). The gas chromatograph was equipped with a Molecular Sieve 5 A capillary column and a packed Carboxen-1000 column. The gaseous products (H_2 , CO, CH_4 , and C_2H_4) were quantified by gas chromatograph

equipped with a flame ionization detector (FID) for CO, CH₄ and C₂H₄, and a thermal conductivity detector (TCD) for H₂. Ultrapure nitrogen (N₂, 99.9999%) was used as the carrier gas. The liquid products (HCOOH and CH₃CH₂OH) were examined by nuclear magnetic resonance spectroscopy (NMR). ¹H NMR and ¹³C NMR spectra were collected on a Bruker Avance III HD 400 NMR spectroscopy in 10% D₂O using water suppression mode, with DMSO as an internal standard.

The Faradaic efficiency (FE) was calculated using the following equation:

$$FE_i = \frac{Q_i}{Q_{total}} = \frac{z_i \times F \times N_i}{I \times t} = \frac{z_i \times \nu_i \times G \times F \times P_o}{I \times R \times T_o \times 60000}$$

where Q_i is the quantity of electric charge needed to produce product i at a certain time interval ($Q_i = z_i \times F \times N_i$); Q_{total} is the quantity of electric charge needed to produce all products at a certain time interval ($Q_{total} = I \times t$); z_i is the number of electrons required to produce one i molecule, which is 2, 2, 8, and 12 for CO, H₂, CH₄, and C₂H₄, respectively; F is the Faradaic constant (96485.3 C mol⁻¹); N_i is the number of moles of product i in the GC sampling loop ($N_i = N_{total} \times V_i$); N_{total} : moles of all gases in the GC sampling loop ($N_{total} = \frac{P_o \times V_o}{R \times T_o}$); V_o : volume of the GC sampling loop ($V_o = G \times t$); I is the measured current; t is the time needed for gas to fill the GC sampling loop; ν_i is the volume ratio of product i obtained from GC; G is the volumetric flow rate; P_o is atmospheric pressure (1.01 × 10⁵ Pa); T_o is the sample loop temperature (298 K); and R is the ideal gas constant (8.314 J mol⁻¹ K⁻¹).

All Faradaic efficiencies and error bars reported in this work were determined from the average of at least three replicates for each potential studied.

Computational methods

All spin-polarized DFT calculations were implemented in the Vienna ab initio Simulation Package (VASP 6.3.0)⁶⁷. The exchange-correlation interactions were modeled by the functional of Perdew, Burke, and Ernzerhof (PBE) within the generalized gradient approximation (GGA), and the projector augmented-wave method was carried out to describe the ion-electron interactions^{68,69}. A Cu (III) (4 × 4) supercell with a lattice constant of 10.22 Å ($\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$) was employed to anchor the Ag atoms or support the Ag₁₃ nanoclusters. The Brillouin zone was sampled by 3 × 3 × 1 k-points by means of Monkhorst-Pack scheme with the cutoff of 400 eV during the structural relaxation, while 5 × 5 × 1 k-points for electronic structure calculation such as projected density of states (PDOS) and charge density difference. The convergence criteria for energy and force were set as 10⁻⁵ eV and 0.01 eV Å⁻¹, respectively. Along the perpendicular direction, a vacuum space at least 18 Å was added to eliminate the effects of periodic structure, and the dispersion corrected density functional theory (DFT-D3) was utilized to include van der Waals (vdWs) interaction between adsorbate and interface⁷⁰.

The adsorption energies of intermediates such as *CO₂, *COOH and *CO were calculated based on the following equation:

$$E_{\text{adsorption}} = E_{\text{total}} - (E^* + E_{\text{adsorbate}})$$

where E_{total} , E^* , and $E_{\text{adsorbate}}$ represent the DFT energy of adsorption system, substrate, and adsorbate, respectively.

Based on the theory proposed by Nørskov et al.⁷¹, the change of Gibbs free energy for each CO₂RR elementary reaction was calculated based on computational hydrogen electrode (CHE) model⁷², referred as $\Delta G = \Delta E + \Delta ZPE - T\Delta S$, where ΔE is the adsorption electronic energy, while ΔZPE and $T\Delta S$ represent zero-point energy correction and entropy contribution at 298 K. For the gas molecules, the entropy was taken from NIST database. For the adsorbed intermediates, only

vibrational entropy was considered, which was calculated from the DFT calculated vibrational frequencies. The activation barrier of each reaction was obtained from climbing-image nudged elastic band (CI-NEB) calculations⁷³.

Data availability

The ex-situ XPS data, ex-situ XAS data, product quantification, *quasi* in-situ XPS data, *operando* XAS data, *operando* Raman data, *operando* ATR-SEIRAS data, isotopic labelling experiment data and theoretical calculations data generated in this study have been deposited in the Figshare database under accession code (<https://doi.org/10.6084/m9.figshare.27199515>)⁷⁴. Source data are provided with this paper.

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Author contributions

S.W., X.L., Y.H., and B.L. designed and conceived the experiment. S.W. performed the catalyst synthesis, structural characterization, and CO₂RR electrocatalytic measurements. F.L. conducted the theoretical calculations. S.W. performed the *operando* electrochemical ATR-

SEIRAS measurements and analyses. Z.L., T.L., S.H., and Y.Lu. contributed to the *operando* XAS spectroscopy measurements and analyses. S.W., Y.Z., and X.R. performed the *operando* Raman spectroscopy measurements and analyses. S.W., Y.Li., and Y.C., performed the *quasi* in-situ XPS measurements and analyses. F.L., Y.C., J.Z. and X.Y. carried out DFT data analysis. W.W. performed the HAADF-STEM measurements and analyses. S.L. performed EMARS method based on frequency filtering over HAADF-STEM images. S.W., X.L., and B.L. wrote and revised the manuscript with inputs from all authors. The project was supervised by X.L., Y.H., and B.L.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Xuning Li, Yanqiang Huang or Bin Liu.

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